

Initiation à la recherche

Etat de l'art

/

Introduction to research

State of the Art

Module « Recherche et séminaires »
ECTS PR422 (Etu) : 2/4 / ECTS (App) : 2/4

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Introduction of the speakers

- **Vincent Berouille**, Course Coordinator
- Teacher/Researcher at Esisar and the LCIS Laboratory since 2002
- PhD in microelectronics from the University of Montpellier
- Director of LCIS
- Research activities focused on “Security and Safety of Integrated Circuits and Systems”

Introduction of the speakers

- **Oum-El-Kheir Aktouf**
- Teacher/Researcher at Esisar and the LCIS Laboratory since 1999
- PhD in computer science from the University of Grenoble
- Former deputy director of LCIS and head of CTSYS team
- Research activities focused on “Safety and security of autonomous and distributed systems”

Preliminary information

- Objective of this chapter:
 - Definition and goals of the “State of the Art”
 - Effective literature search (tools, databases, methods)
 - Critical analysis of sources (different types of sources, predatory journals)
 - Synthesizing information about the research question

Preliminary information

- Evaluation method:
 - Phase 1 (lectures):
 - Multiple-choice quiz (QCM): 10%
 - Phase 2 (« State of the art » project):
 - Milestone meeting: 10%
 - State of the art document + analysis: 30%
 - Oral defense: 50%

Outline

- Project introduction
- Objectives and definitions
- Documentary research
- Search techniques
- Processing research results
- Bibliographic referencing

Project introduction

- Objectives
- Schedule
- Evaluation
- Student time management

Project introduction

Objectives

- Write a State of the Art:
 - on a specific « research question »
 - 2 to 4 pages, double columns, IEEE format, including references
 - Oral presentation 5-10mn on the overall organization of the paper followed by 15mn specific questions for each student on the paper
- Optionnal: Systematic literature review method (1 page)
- Team work (4-5 students per team)
 - Team formation based on ranked individual preferences

Project introduction

Schedule PR422

- Start: 5 October
- Milestone: week of 2 November
- Detailed outline deadline : 20 November
- Final document: 18 December
- Defense: January

Project introduction

Schedule PR423

- Start: 20 April
- Milestone deadline: 7 May
- Detailed outline deadline: 21 May
- Final document deadline: 3 June
- Defense: 11 June

Project introduction

Evaluation

- Milestone presentation: 10%
- Final document: 40%
- Final oral presentation: 40%
- Optional: meta-analysis (keywords, date, conferences, authors, etc)

Project introduction

Milestone presentation - expectations

- Two main aspects are evaluated:
 - Structure of the paper and team organization: each section must associated to one student
- Work expected from groups:
 - 3–4 relevant references per student
 - State of the art divided into sections with associated references
 - Each student assigned to a specific section (plus introduction and/or conclusion)
- Organization:
 - Meeting schedule to be arranged by tutors with their student groups

Project introduction

Final document

- Scientific Content and Analysis
 - Clearly present the research context and problem statement
 - Define all terms and abbreviations
 - Avoid too many details: focus on comparison between papers and **HIGH LEVEL INFORMATION**
 - Define common comparison criteria
 - Cite references explicitly instead of using “we”
 - Prefer clarity and simplicity over exhaustive detail
 - **AI can be used but: Be able to explain every sentence/slide**
 - Be careful with automatic translation and false friends

Project introduction

Oral presentation (French or English)

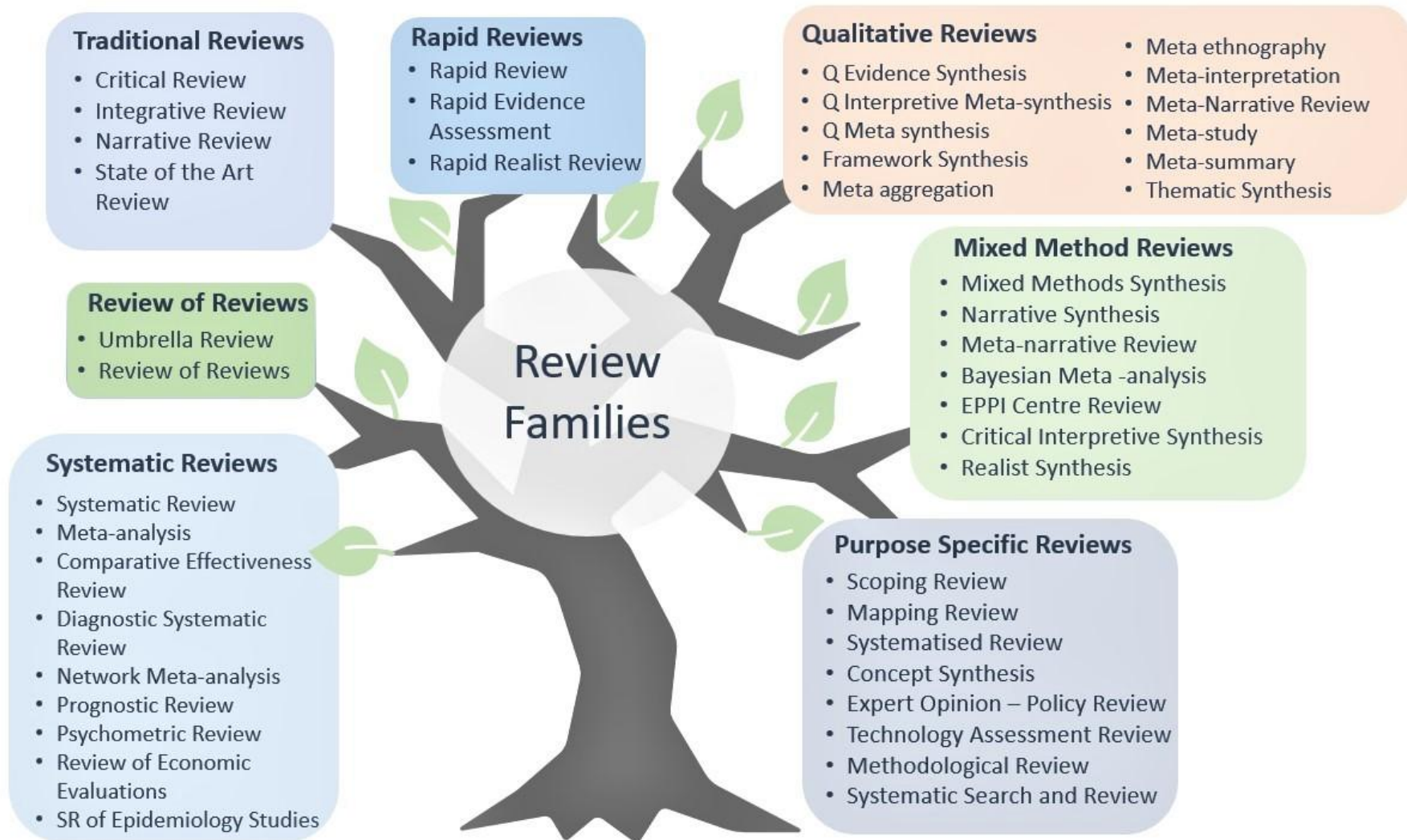
- Presentation Organization and Delivery
 - Add slide numbering
 - Respect the presentation allocated time (+-10% approximately)
 - Ensure balanced timing between presenters
 - Briefly introduce the presenters
 - Do not write too much text in the slides, but use figures and cite references if you reuse figures
 - **Do not use a phone or tablet (Use a sheet of paper for notes but do not read from it constantly)**
 - Respect submission deadlines
 - Do not hesitate to ask your supervisor for feedback
 - AI can be used but: Be ready to answer questions related to your paper section

Project introduction

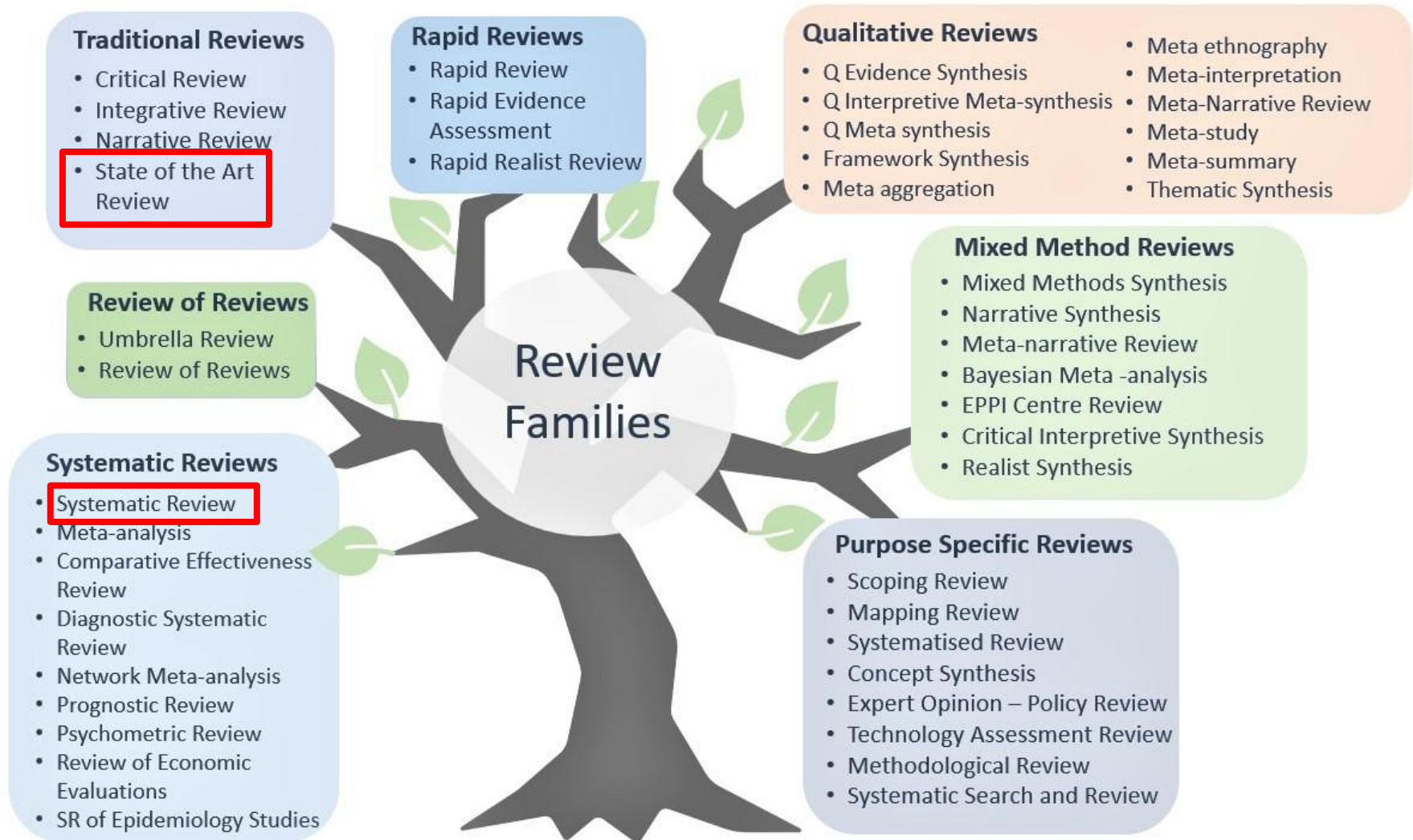
Student time management

- 1h meeting with tutor
- 6h research and preliminary reading of papers
- 3h paper organization & task distribution (**milestone prep**)
- 1h milestone meeting
- 9h in-depth reading of papers (**3-4 per student**)
- 1h writing a detailed outline of each part
- 3h writing the section
- 2h proofreading and coordinating introduction/conclusion
- 3h preparation for the defense

What is a review study?



What is a review study?

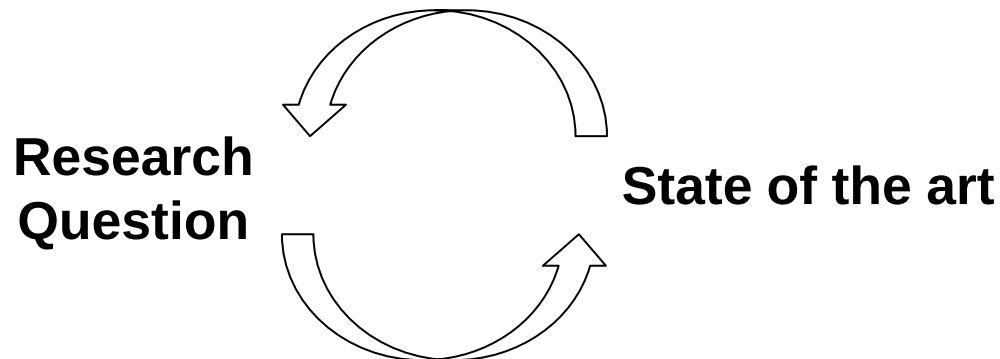


Objectives and definitions

- The « *State of the Art* » is a focused synthesis of the current best knowledge and techniques on a **research question**
 - Often based on a literature review, but with a **problem-solving orientation**
 - Highlights limits, gaps, and open questions
 - **Recognizing the prior work** on a subject
 - Avoids unnecessary duplication of work and builds upon existing knowledge

Objectives and definitions

- **Definition of a “research question”:**
 - *a precise, focused question identifying an unknown in existing knowledge*
 - Helps narrow down the information research (if it's too broad, Wikipedia might suffice).
 - Often the state of the art **helps refine the research question.**



Objectives and definitions

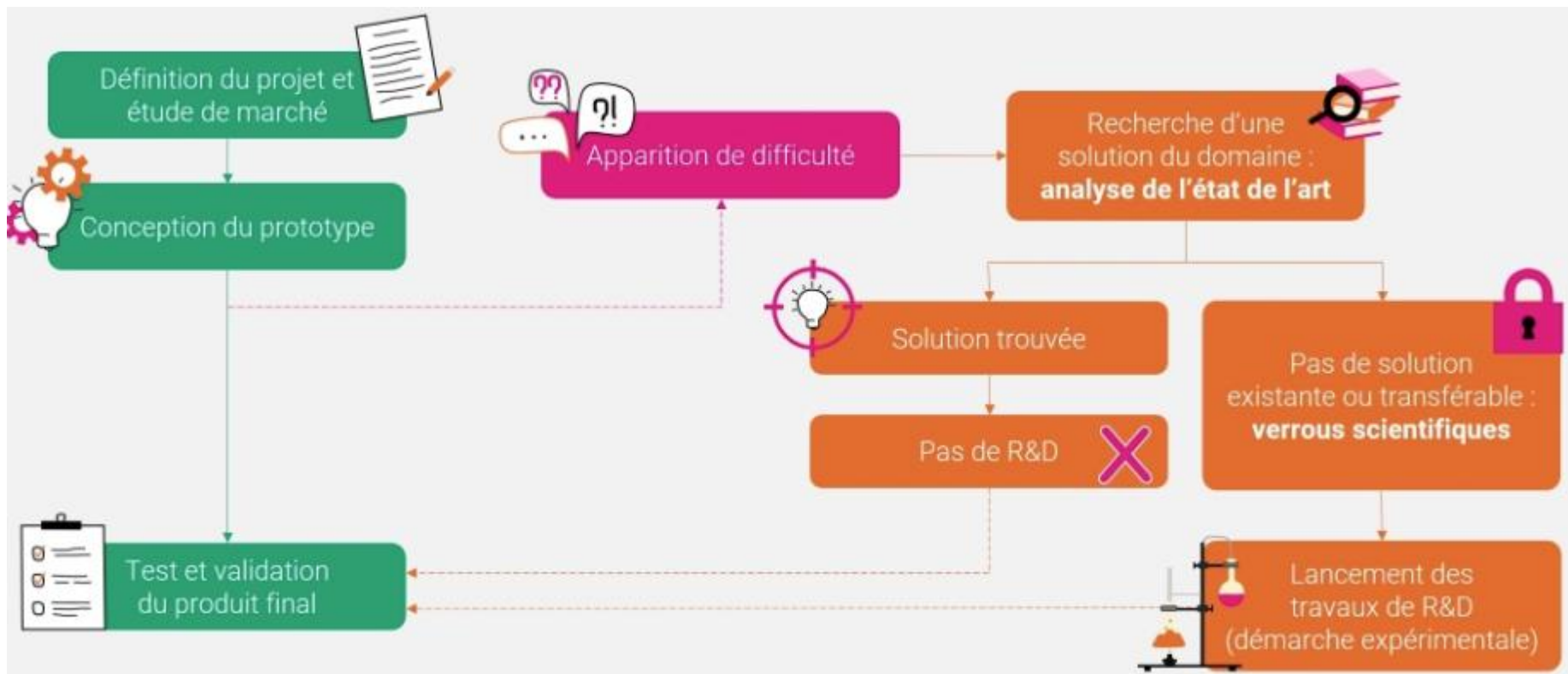
- The *State of the Art* is more than a “literature review” or “bibliographic research” (which are “*all the steps involved in searching, identifying, and finding documents related to a topic*”)
- The *State of the Art* is **not a market study, but a review of existing scientific knowledge**

Objectives and definitions

- Examples of research questions:
 - How effective are glitch detection techniques against voltage fault attacks on cipher blocks in submicronic technologies?
 - How do process variations impact the reliability of SRAM cells in modern technologies?
 - What is the impact of power amplifier non-linearity on QAM signal integrity in IoT devices?
 - What is the accuracy of neural network-based system identification for non-linear dynamic systems?

Objectives and definitions

- Often a required step for asking the **CIR (Crédit Impôt Recherche)**.



Documentary Research

- Goal: Find relevant information and process it.
- New technologies = allow processing huge amount of information.
- Bibliographic databases: Scopus, Springer, ScienceDirect, IEEE (not free but accessible via university libraries)

Scopus®

IEEE Xplore®
Digital Library



Springer

Documentary Research

- Open archives (HAL et arXiv) **accessible to everyone for free**



Documentary Research

- Specialized search engine **Google Scholar** (not Google!) indexes scientific literature, bibliographic databases, and open archives.

The logo for Google Scholar. The word "Google" is in its characteristic multi-colored font (blue, red, yellow, blue, green, red), and the word "Scholar" is in a plain grey sans-serif font.

<https://scholar-google-com.sid2nomade-1.grenet.fr/>

Search Techniques

- Choose the right keywords = keywords evolve while reading (check keywords in good papers).
- Use Boolean operators like in Google (AND, OR, NOT, quotation marks, + filetype:pdf).

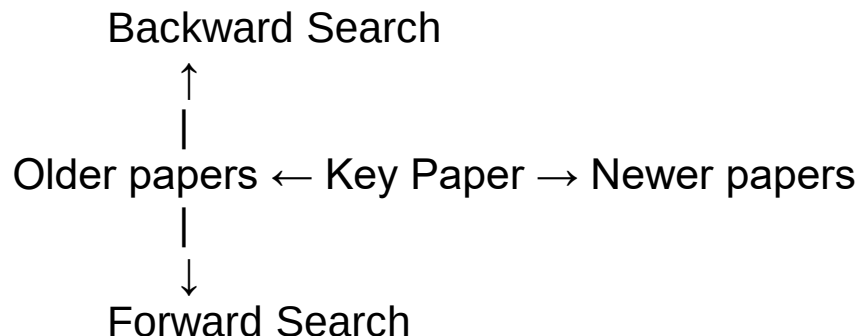
Search Techniques

- Read references from good papers and continue the search, also check which papers cite them, and look up other publications by their authors.
- **Some papers are surveys or reviews = big time saver and good starting points.**

Search Techniques

Forward & Backward Search

- **Backward Search**
 - Look at the **references cited by a key paper**
 - Find **earlier foundational works**
- **Forward Search**
 - Find **recent papers that cite the key paper**
 - Tracks the **latest developments and progress**



Search Techniques

- Pay attention to publication dates: **too old?**
- Check publication quality:
 - [conference ranking sites](#) (often based on conference h-index)
 - Journal ranking sites ([Scimago](#))
 - Distinguish by type: PhD or Master thesis, workshop, symposium, conference, journal (peer-review level can be different and affects accuracy and reliability of results).

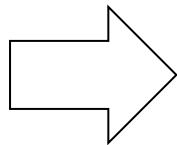
Search Techniques

- Be careful about exhaustiveness = avoid missing key papers.
- A **systematic literature review** is a **rigorous and structured method** to thoroughly explore a research topic
- It's not just “I read what I find” — it's “**I decide how I'll search and analyze before starting**”

Search Techniques

- **systematic literature review method:**

- 1. Set search strategy** (databases, keywords, period)
- 2. Apply inclusion & exclusion criteria**
- 3. Select relevant papers**



reproducible

Processing research results

- Mainly read abstract, then introduction, then conclusion.
- Classify papers by categories (mind map):
 - a good bibliography proposes a categorized organization of results
 - helpful for writing the state of the art and structuring future work.

Processing research results

- **Classify papers by theme**
 - Introduce each theme and explain its importance
- **For each paper:**
 - In [Author, Year], a new method is proposed to ...Based on ... However, this approach does not allow ...
- **General Introduction**
 - Context of the study
 - Research questions
 - Paper organization
- **General Conclusion**
 - Summarize key findings and limitations

Processing research results

- For each paper, identify gaps relative to your focus : *“this study does that but in a different context”*
- List possible contradictions or divergences among references
- A state of the art includes **no value judgments or personal information.**

Processing research results

- **Bibliographic referencing:** standardized and depends on publishers.
- Tools available to organize/manage references: **Mendeley**, **Zotero** (open source), **EndNote** (not free).

Bibliographic Management Tools

— Comparison

Criteria	Zotero	Mendeley	EndNote
License	Free, open source	Free (with paid options)	Paid (academic license available)
Platforms	Windows, Mac, Linux, Web	Windows, Mac, Web	Windows, Mac
Cloud / Sync	Yes (free limited storage)	Yes (free limited storage)	Yes (depending on license)
Word Integration	Yes (plugin)	Yes (plugin)	Yes (native)
Ease of Use	Very easy	Easy	More technical / advanced
Collaboration	Yes (shared groups)	Yes (shared groups)	Possible (depends on version)
Export Formats	BibTeX, RIS, EndNote, etc.	BibTeX, RIS, EndNote, etc.	Wide range
Editor Independence	Yes	No (owned by Elsevier)	No (owned by Clarivate Analytics)
Typical Usage	Students, PhD candidates, researchers	Technical and medical sciences	Medicine, life sciences, pro publishing

Processing research results

- **Zotero** is increasingly preferred in academic research for its simplicity, openness, and independence.

zotero

Processing research results

References

- Several standards exist: **IEEE, APA, MLA, Chicago...**
- In **scientific and technical fields**, **IEEE** is the **most widely used**.
- Used in conferences, journals, and research papers in **engineering, electronics, and computing**.

Processing research results

References

- **Format:**

[N°] Initial(s). Last name, "Title of the article," *Journal name*, vol., no., pp., month year

- **Example:**

[1] J. Dupont, "A fault injection technique for secure ICs," *IEEE Transactions on Computers*, vol. 71, no. 5, pp. 123-135, May 2024.

Processing research results

References

- Use **[x]** to cite references.
- **x increases in order of appearance** in the text.
- Follow the **IEEE citation style**.
- Example:
Several fault injection techniques have been proposed to bypass cryptographic protections in ICs [1]-[2]. A recent study improved detection accuracy for voltage glitches [3].

Processing research results

References

Case	Correct	Incorrect
Single reference	[5]	[5] (extra space)
Two consecutive references	[5]–[6]	[5-6] or [5][6]
Multiple consecutive references	[5]–[9]	[5][6][7][8][9]
Two non-consecutive references	[5], [7]	[5 7]
Mixed (consecutive + isolated)	[5]–[7], [10], [12]	[5-7,10,12]

References

[1] <https://unimelb.libguides.com/whichreview/stateoftheartreview>

[2] S. Keshav. “How to Read a Paper”, available at
<https://web.stanford.edu/class/ee384m/Handouts/HowtoReadPaper.pdf>